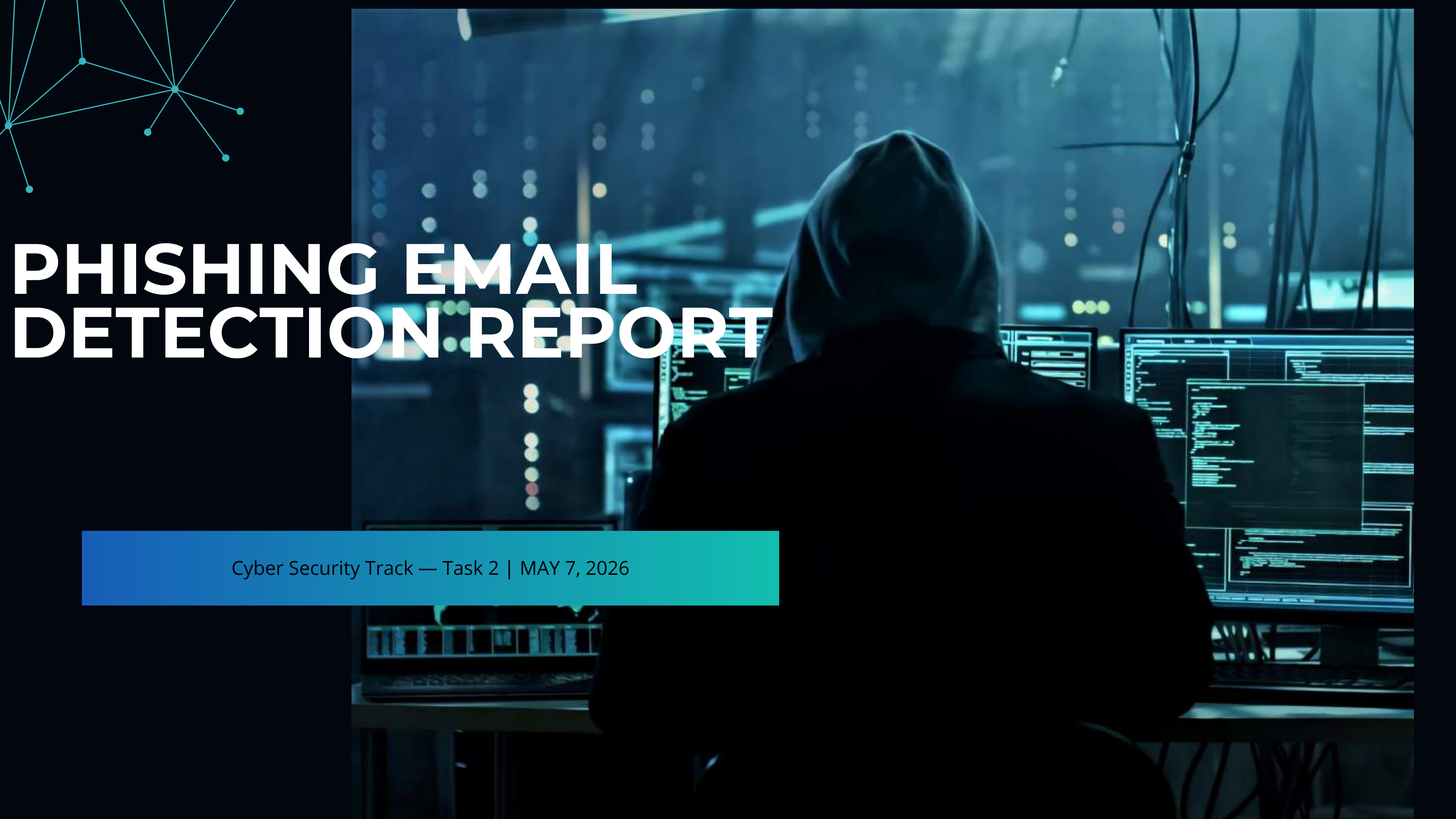




PHISHING EMAIL DETECTION REPORT

Cyber Security Track — Task 2 | MAY 7, 2026



EXECUTIVE SUMMARY

3 phishing emails were analyzed using MXToolbox and Browser DevTools. All 3 confirmed as PHISHING.

Targets: Bank of America, PayPal, Microsoft

Total Red Flags Found: 20+

Overall Risk Rating: HIGH



METHODOLOGY & TOOLS USED

- MXToolbox — Email header analysis (SPF, DKIM, blacklisted IPs)
- Browser DevTools — Safe URL inspection without clicking suspicious links
- Manual Analysis — Sender domain review, link inspection, language analysis

EMAIL CLASSIFICATION SYSTEM



● SAFE Passes SPF/DKIM, legitimate domain



● SUSPICIOUS Some warning signs, not confirmed



● PHISHING Fails authentication, spoofed domain, malicious links, urgency tactics

All 3 samples = PHISHING

SAMPLE 1 — FAKE BANK OF AMERICA

FROM: SECURITY@BANKOFAMERICA-ALERT.COM
SUBJECT: URGENT: YOUR ACCOUNT HAS BEEN SUSPENDED

- Red Flags: - Fake domain — bankofamerica-alert.com
- SPF: FAIL | DKIM: NONE
 - Malicious link — secure-login.xyz
 - Urgency: 24 hours or account closed
 - Generic greeting — Dear Valued Customer

VERDICT: PHISHING



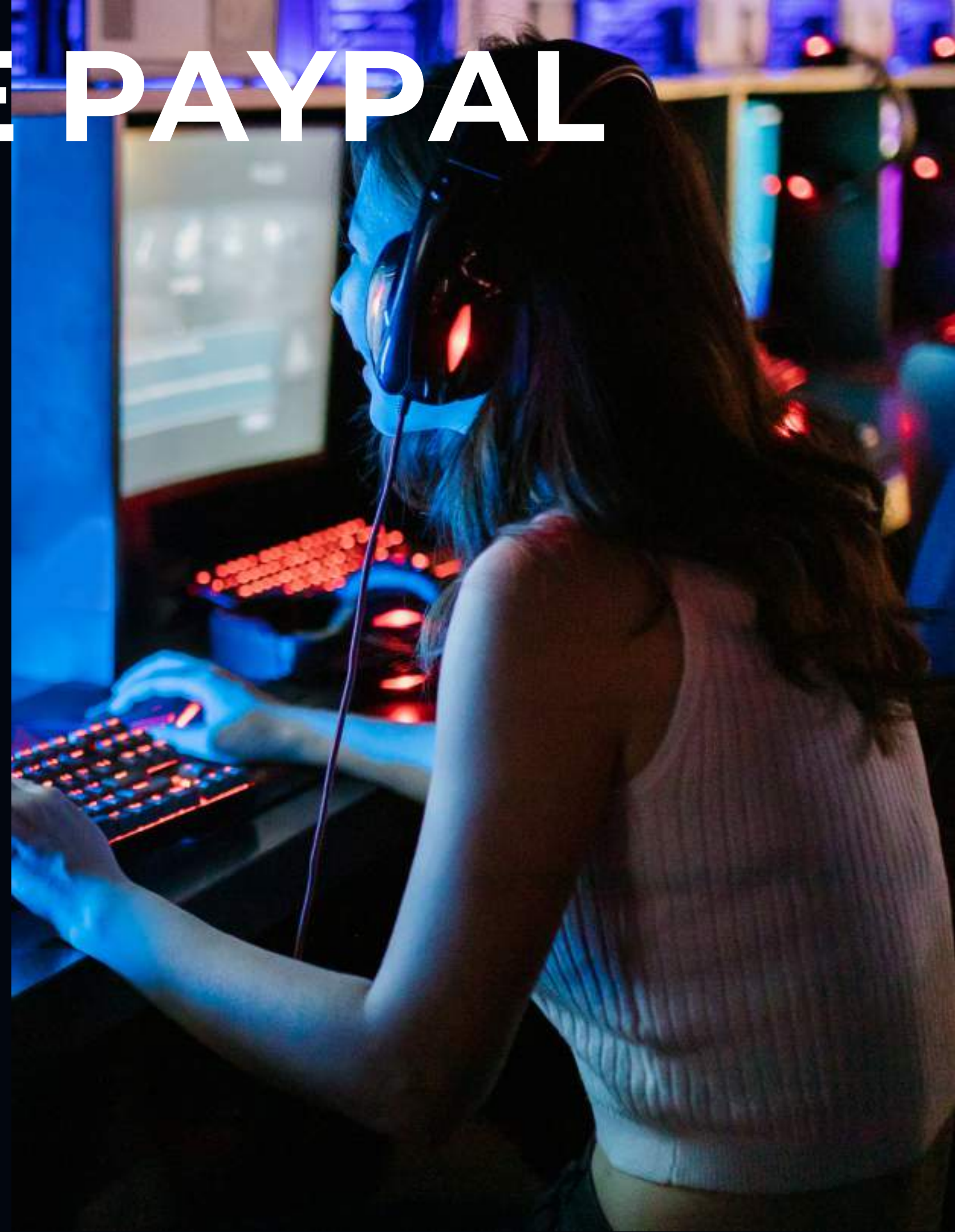
SAMPLE 2 — FAKE PAYPAL ALERT

FROM: SERVICE@PAYPAL-SUPPORT.NET
SUBJECT: YOUR PAYPAL ACCOUNT IS LIMITED

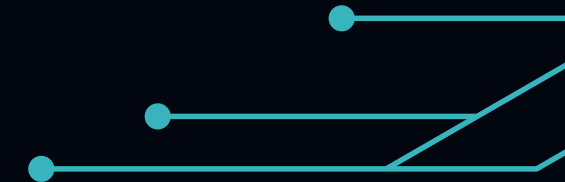
Red Flags:

- Fake domain — paypal-support.net
- SPF: FAIL | DKIM: NONE
- IP 185.220.101.45 — BLACKLISTED
- Malicious link — account-verify.ru
- X-Mailer: The Bat! 9.3 (spam tool)

VERDICT: PHISHING — Most dangerous!



SAMPLE 3 — FAKE MICROSOFT ALERT



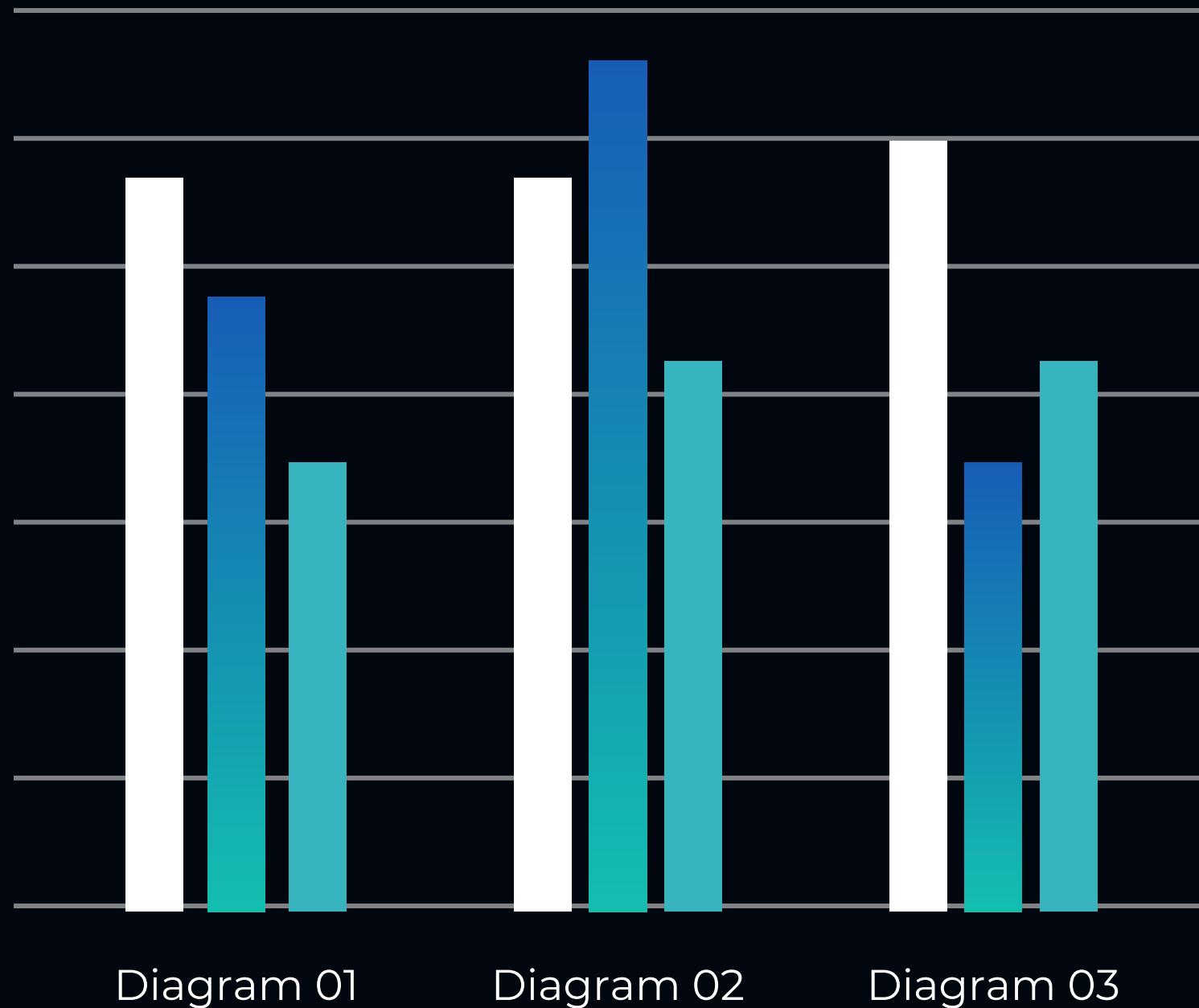
FROM: NO-REPLY@MICROSOFT-SECURITY.INFO
**SUBJECT: ACTION REQUIRED: SIGN-IN ATTEMPT
BLOCKED**

Red Flags:

- Fake domain — microsoft-security.info
- SPF: FAIL | DKIM: NONE
- Malicious link — account-check.com
- Urgency: account will be locked
- X-Mailer: PhpMailer 6.0 (bulk script)

VERDICT: PHISHING — Targets employees!

PHISHING TECHNIQUES IDENTIFIED



Compliance Metrics

- Domain Spoofing — Fake domains mimicking real brands
- SPF Failure — Unauthorized sending server
- Missing DKIM — No digital email signature
- Urgency Tactics — 24hrs, account closure threats
- Lookalike URLs — paypal.com.verify.ru
- Generic Greetings — Dear Customer
- Blacklisted IPs — Known malicious servers

PREVENTION & AWARENESS GUIDELINES



Before opening emails:

- Check sender domain after @ symbol
- Look for your real name in greeting
- Be suspicious of urgent threats



Before clicking links:

- Hover to see real URL destination
- Go directly to website instead



If you receive phishing:

- Do NOT click any links
- Report to IT team immediately
- Delete from inbox and trash

FINAL SUMMARY & CONCLUSION

Sample 1 — Fake Bank of America → PHISHING

Sample 2 — Fake PayPal → PHISHING (IP Blacklisted)

Sample 3 — Fake Microsoft → PHISHING

All 3 failed SPF & DKIM authentication. All 3 used domain spoofing and fear tactics.

Recommendation: Implement SPF, DKIM, DMARC and conduct regular phishing awareness training.



Assessed by: Evans Njenga | MAY 7, 2026